Wei-Jhih Huang

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Research Interests

AI-Native 6G Networks, Learning-based Wireless Systems, Radio Resource Management, Mobile and Edge Computing

Education

Bachelor of Science, Communication Engineering, National Central University **Sep 2022 – Jun 2026**
GPA: 3.94/4.3 Taoyuan, Taiwan
Relevant Courses: *Communication Theory, Digital Signal Processing, Data Structures, Algorithms, Probability, Computer Networks, Protocols and Applications for Wireless Sensor Networks, and Programming for Deep Learning*

Research Experience

Wireless Mobile Network Lab, National Taiwan University **Nov 2026 – Present**
Research Assistant, Advisor: **Prof. Hung-Yu Wei** Taipei, Taiwan

- Investigating radio link failure prediction for high-mobility 5G metro networks that combines short-term, physics-based radio indicators with long-term, learning-based predictors, using real-world 5G measurement data.
- Exploring DRL-based cross-layer optimization for AI-native RAN that jointly orchestrates radio resources and edge LLM inference, keeping AI services responsive in high-mobility transit environments by acting on predicted connectivity disruptions.

Information Processing and Communications Lab, National Central University **Feb 2024 – Jun 2026**
Undergraduate Researcher, Advisor: **Prof. Chih-Wei Huang** Taoyuan, Taiwan

- Proposed a reward-agnostic fine-tuning trigger for DRL-based radio resource management using VAE trajectory embeddings and Jensen-Shannon divergence, improving OOD detection accuracy by ***16.7%*** and UE satisfaction by up to ***11%*** on unseen traffic.
- Co-developed PACI-Net, a physics-aware distillation framework converting black-box DRL schedulers into interpretable students via 3GPP-aligned concepts and pairwise interactions, improving physical-concept correlation by ***23.9%*** over the strongest baseline.
- Explored a self-aware meta-RL framework for 6G radio resource management that detects traffic distribution shifts via a feature-wise attention gate and latent-space OOD detection, triggering targeted model fine-tuning.

Electronic Design Automation Lab, The Chinese University of Hong Kong **Jul – Aug 2025**
Summer Visiting Research Student, Advisor: **Prof. Bei Yu** Hong Kong

- Contributed to NUA-Timer++, a graph-based framework for hierarchical RTL timing prediction under non-uniform input arrival times, which achieves a 1.96x improvement in consistency and reduces average error to ***0.42x*** compared to the state of the art.
- Evaluated Transformer architectures and attention mechanisms for reasoning over candidate critical paths, and independently investigated a critical-path-based timing constraint extension as an auxiliary signal for logic synthesis.

Publications

- **Wei-Jhih Huang**, Jin-Min Yang, Ning Ma, Sheng-Liang Wu, Chih-Wei Huang. "Deviation-Aware Trajectory Embedding for Fine-Tuning Triggering in DRL-Based Wireless Network Resource Management," *IEEE Wireless Communications and Networking Conference (WCNC),* April 2026, https://doi.org/10.1109/WCNC65185.2026.11555618.
- Ching-Wei Lin, **Wei-Jhih Huang**, Sheng-Liang Wu, Chih-Wei Huang, Phone Lin, Shun-Ren Yang. "PACI-Net: Physics-Aware Distillation for Explainable Radio Resource Management," *IEEE Global Communications Conference (GLOBECOM),* December 2026 (to appear).
- Ziyi Wang, **Wei-Jhih Huang**, Fangzhou Liu, Tsung-Yi Ho, David Z. Pan, Bei Yu. "NUA-Timer++: Timing Prediction for Hierarchical RTL under Non-Uniform Input Arrival Times," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2026 (to appear).

Honors & Awards

- **Undergraduate Research Grant,** National Science and Technology Council, Taiwan, 2025–2026
- **Outstanding Student Scholarship,** National Central University, 2026
- **Outstanding College Youth Award,** School Level, Taiwan, 2026

- **Honorable Mention,** Undergraduate Project Competition, Department of Communication Engineering, NCU, 2025
- **Service-Learning Honors,** Gold & Merit Awards, Public Service Scholarship, Club Excellence Award, NCU, 2024–2026

Teaching Experience

National Central University **Feb – Jun 2025**

Teaching Assistant, Courses: ***Natural Language Processing*** *and* ***Enterprise Application of Pattern Recognition***

- Assisted in lecture preparation and grading for NLP topics and image recognition topics for ***120*** students, analyzing the underlying AI technologies of real-world company products to support students in final project research.

Work Experience

Chunghwa Telecom **Sep – Dec 2025**

AI Software Development Semester Intern Taipei, Taiwan

- Spearheaded the full-stack development of an intelligent project management system, spanning UI/UX design and n8n workflow backend orchestration, driven by RAG-based agentic models to enhance operational efficiency by ***33%.***
- Selected to represent the department at the *2025 IEEE Global Communications Conference*, introducing the department's AI-powered intelligent customer service products to international industry and academic professionals.

Industrial Technology Research Institute **Jun – Oct 2024**

AI Software Development Intern Hsinchu, Taiwan

- Built and deployed an LLM-based service chatbot at *Taipei Songshan Sports Center*, integrating Azure OpenAI with Linux backend services to provide ***24/7*** automated customer support, reducing customer queue time by ***50%*** and improving service efficiency.
- Selected to demonstrate the department's AI chatbot at the *2024 Smart Healthcare Summit* and *2024 Healthy Ageing Tech Show*.

Skills

Machine Learning: Deep Reinforcement Learning, Deep Generative Models, Graph Neural Networks, Transformers
AI Applications: RAG pipeline, Agentic workflow (n8n), LLM-based chatbot, Full-stack AI integration
Programming: Python (PyTorch), C/C++, JavaScript (Node.js, Express, React)
Systems & Deployment: Linux, Docker, Azure, REST APIs, Git, OpenAI API
Languages: English (Professional Proficiency), Mandarin Chinese (Native)

Leadership & Extracurriculars

NCU Rural Service Club **Feb 2023 – Jul 2025**

Club President & Service Team's Group Leader

- Secured multiple awards and scholarships as president and reinvested ***60%*** of prize funds into club equipment and team-building.
- Led and coordinated educational camps for ***160*** children across remote elementary schools in Yilan in February 2023 and Taichung in January 2024, designing interactive lessons and managing team activities to foster creativity and learning.

NCU Wireless Communication and Network Multimedia Camp **Jul 2023**

AI Instructor & Course Team Member

- Designed and taught an introductory ML and LLM curriculum, making core AI concepts accessible to ***70*** high school students.
- Guided students in completing hands-on Arduino projects to foster practical understanding of wireless communication.